Coding with Jiangly

2.7

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Problems are written by Codeforces writers. Codeforces platform is run by MikeMirzayanov. Submissions are by jiangly, and the book is collated by 2.7.

Introduction

There are many good players at codeforces. Like many people, I read their submissions for inspirations and practice helps. Among them, jiangly stands out for having one of the most readable code.

Benq and Um_nik use a large template, and tourist also fall in that category. Jiangly, however, only copies templates when needed, and generally writes clean code. Personally, I find it much easier to learn from shorter, more-readable submissions. I first had the idea for this project in Jan21, when jiangly just became an LGM, and it is great to see jiangly climb to be one of the best players. While I don't know about the history of the jiangly fan club, perhaps that is why so many of us are there.

I took the first 50 pages of jiangly's submissions, combined them with tags / other information, and put them together as a large problem set. I suspect it is unnecessary to try all the problems at a level when trying to improve, given the sheer quantity of problems here.

The difficulty ranges are divided into the following, roughly matching the rank colors. Not including 3000+ because of a coding bug.

1. Sub-1200 (Gray)
2. 1250-1600 (Green)

3. 1650-1900 (Blue)
4. 1950-2100 (Violet)
5. 2150-2300 (Orange)
6. 2350-2600 (Rose)
7. 2650-3000 (Red)

For each range, I sorted the tags by frequency, and selected the tags with the most frequent appearences. I chose these until the tags account for at least 70% of all tags by frequency. For each I currently included the link to the problem and a link to jiangly's solution. At the end of each section I also included an assortment of 20 problems, randomly selected from the remaining ones.

What might make this a good selection of problems? I think there are two common approaches to practicing problems, namely solving contests & solving handout problems. Both have their pros and cons.

1. Solving contests gets you an *accurate* sample of problems you will see in contexts, but does not give you the right level. A 2k-rated player will encounter both 3k-rated and 800-rated problems in a typical contest, and neither is very educational.
2. Handout problems can be tuned to be the right level. However, they often focus on interesting, but infrequent topics. You will find many exotic data structures browsing through Codeforces. The less flashy techniques (e.g. Greedy and DP), however, make up the majority of the problems.

I hope the selected problems can get the *best of both worlds*. The chapters are divided by difficulty, so that one can find problems that are at the right level. The sections of a chapter are selected using the most common tags found on Codeforces, so they should be teaching the most important techniques. Of course, I didn't include every problem of a given tag, since solving 10-12 of each is already plenty of work.

Why do these here, instead of from the problemset page? I think solving from the problemset page is a great approach as well, but this might still be an improvement. 1. It is easier to schedule practice with a book. One can say "solve problems 430-440 from the blue book" and you will easily know where to find them. 2. Each of these problems contain a good selected solution. The solution here is answering "how would a black-level player solve this under time pressure?", and are perhaps more practical than the curated solutions. They will not be perfect, but they should be good. 3. It saves you the hassle of finding problems.

What is up with the struct's? There are some code that are frequently repeated, such as that for modular arithmetic. Including them tends to make the code look less neat, so I took them out. The common ones are found in the appendix.

What does this book not have? Of course, given I compiled this using a program, there are a few limitations. 1. Explanation of the concepts. You can look around Codeforces, wikipedia, or ask an AI to

explain it. 2. Quality control. If a problem is way too easy / brutal for its difficulty, or has nothing to do with the tag, feel free to comment it.

Finally, an observation: if you can solve all the problems in this book without looking at hints, in (30 min for hard ones, 5min for easy ones), your tournament performances should be almost indistinguishable from his. There are some notable differences, of course: 1. You know that some black-level player, namely jiangly, was able to solve this problem. Jiangly may not have known that. 2. You know the tags that the problem received.

Now, enjoy the problems!

gray

brute force

1: Erase First or Second Letter

Problem: link

Solution: link

2: Easy As ABC

Problem: link

Solution: link

3: Don't Try to Count

Problem: link

Tutorial: link

Solution: link

4: Grandma Capa Knits a Scarf

Problem: link

Tutorial: link

Solution: link

5: Two Vessels

Problem: link

Tutorial: link

Solution: link

6: Download More RAM

Problem: link

Solution: link

7: Cirno's Perfect Bitmasks Classroom

Problem: link

Solution: link

8: Pizza Separation

Problem: link

Tutorial: link

Solution: link

9: Beautiful Divisors

Problem: link

Tutorial: link

Solution: link

10: Fancy Coins

Problem: link

Tutorial: link

Solution: link

11: ACM ICPC

Problem: link

Solution: link

12: Local Extrema

Problem: link

Tutorial: link

Solution: link

13: Another Permutation Problem

Problem: link

Solution: link

14: Fibonaccharsis

Problem: link

Tutorial: link

Solution: link

15: Desorting

Problem: link

Tutorial: link

Solution: link

16: Longest Divisors Interval

Problem: link

Solution: link

17: Balanced Round

Problem: link

Solution: link

18: Rudolph and Tic-Tac-Toe

Problem: link

Solution: link

19: Trust Nobody

Problem: link

Tutorial: link

Solution: link

20: Gold Rush

Problem: link

Solution: link

constructive algorithms

21: Binary Imbalance

Problem: link

Tutorial: link

Solution: link

22: Sorting with Twos

Problem: link

Solution: link

23: Chips on the Board

Problem: link

Tutorial: link

Solution: link

24: MEXanized Array

Problem: link

Tutorial: link

Solution: link

25: Make It Zero

Problem: link

Solution: link

26: XOR Palindromes

Problem: link

Solution: link

27: green_gold_dog, array and permutation

Problem: link

Solution: link

28: Madoka and the Elegant Gift

Problem: link

Tutorial: link

Solution: link

29: Patchouli's Magical Talisman

Problem: link

Solution: link

30: Difference of GCDs

Problem: link

Solution: link

31: Two Binary Strings

Problem: link

Solution: link

32: Prime Deletion

Problem: link

Solution: link

33: Swap and Reverse

Problem: link

Solution: link

34: Increasing and Decreasing

Problem: link

Solution: link

35: Sequence Game

Problem: link

Tutorial: link

Solution: link

36: Not a Substring

Problem: link

Tutorial: link

Solution: link

37: Yet Another Permutation Problem

Problem: link

Solution: link

38: Olya and Game with Arrays

Problem: link

Solution: link

39: United We Stand

Problem: link

Solution: link

40: Escalator Conversations

Problem: link

Tutorial: link

Solution: link

greedy

41: Training Before the Olympiad

Problem: link

Tutorial: link

Solution: link

42: Unnatural Language Processing

Problem: link

Solution: link

43: Three Activities

Problem: link

Tutorial: link

Solution: link

44: Quests

Problem: link

Tutorial: link

Solution: link

45: Begginer's Zelda

Problem: link

Solution: link

46: Removal of Unattractive Pairs

Problem: link

Tutorial: link

Solution: link

47: Chip and Ribbon

Problem: link

Tutorial: link

Solution: link

48: Line Trip

Problem: link

Tutorial: link

Solution: link

49: StORage room

Problem: link

Solution: link

50: Halloumi Boxes

Problem: link

Solution: link

51: Getting Points

Problem: link

Tutorial: link

Solution: link

52: Cover in Water

Problem: link

Solution: link

53: Yarik and Array

Problem: link

Tutorial: link

Solution: link

54: Points and Minimum Distance

Problem: link

Tutorial: link

Solution: link

55: Haunted House

Problem: link

Tutorial: link

Solution: link

56: Simple Design

Problem: link

Tutorial: link

Solution: link

57: Gamer Hemose

Problem: link

Solution: link

58: Make Them Equal

Problem: link

Solution: link

59: Helmets in Night Light

Problem: link

Solution: link

60: How Much Does Daytona Cost?

Problem: link

Solution: link

implementation

61: 2023

Problem: link

Tutorial: link

Solution: link

62: Can I Square?

Problem: link

Solution: link

63: Not Quite Latin Square

Problem: link

Solution: link

64: Odd One Out

Problem: link

Solution: link

65: Distinct Buttons

Problem: link

Solution: link

66: Problemsolving Log

Problem: link

Tutorial: link

Solution: link

67: Rating Increase

Problem: link

Tutorial: link

Solution: link

68: YetnotherrokenKeoard

Problem: link

Tutorial: link

Solution: link

69: Rook

Problem: link

Tutorial: link

Solution: link

70: 250 Thousand Tons of TNT

Problem: link

Tutorial: link

Solution: link

71: Milica and String

Problem: link

Solution: link

72: Perfect Square

Problem: link

Tutorial: link

Solution: link

73: 1D Eraser

Problem: link

Solution: link

74: Target Practice

Problem: link

Solution: link

75: Short Sort

Problem: link

Solution: link

76: Divine Array

Problem: link

Solution: link

77: Two Subsequences

Problem: link

Solution: link

78: The Corridor or There and Back Again

Problem: link

Tutorial: link

Solution: link

79: Life of a Flower

Problem: link

Tutorial: link

Solution: link

80: Game

Problem: link

Tutorial: link

Solution: link

math

81: Two Divisors

Problem: link

Tutorial: link

Solution: link

82: Least Product

Problem: link

Solution: link

83: Make Almost Equal With Mod

Problem: link

Solution: link

84: Preparing for the Contest

Problem: link

Tutorial: link

Solution: link

85: Constructive Problems

Problem: link

Solution: link

86: Laura and Operations

Problem: link

Solution: link

87: Game with Integers

Problem: link

Tutorial: link

Solution: link

88: Treasure Chest

Problem: link

Tutorial: link

Solution: link

89: Deja Vu

Problem: link

Solution: link

90: Hemose Shopping

Problem: link

Solution: link

91: Special Numbers

Problem: link

Solution: link

92: Consecutive Sum Riddle

Problem: link

Solution: link

93: Three Threadlets

Problem: link

Tutorial: link

Solution: link

94: Fear of the Dark

Problem: link

Tutorial: link

Solution: link

95: Sum of Three

Problem: link

Tutorial: link

Solution: link

96: Joyboard

Problem: link

Solution: link

97: Goals of Victory

Problem: link

Solution: link

98: Vasilije in Cacak

Problem: link

Solution: link

99: Aleksa and Stack

Problem: link

Solution: link

100: Good Kid

Problem: link

Solution: link

misc

101: LOL Lovers

Problem: link

Tutorial: link

Solution: link

102: Swap and Delete

Problem: link

Tutorial: link

Solution: link

103: Increasing Sequence

Problem: link

Solution: link

104: Rigged!

Problem: link

Tutorial: link

Solution: link

105: Friendly Arrays

Problem: link

Tutorial: link

Solution: link

106: Luntik and Subsequences

Problem: link

Tutorial: link

Solution: link

107: Luntik and Concerts

Problem: link

Tutorial: link

Solution: link

109: Era

Problem: link

Solution: link

110: A.M. Deviation

Problem: link

Tutorial: link

Solution: link

111: Coloring Rectangles

Problem: link

Tutorial: link

Solution: link

112: Mathematical Addition

Problem: link

Tutorial: link

Solution: link

113: Plus Minus Permutation

Problem: link

Tutorial: link

Solution: link

114: Non-coprime Split

Problem: link

Tutorial: link

Solution: link

115: GCD Arrays

Problem: link

Solution: link

116: Fun with Even Subarrays

Problem: link

Solution: link

117: Min Max Swap

Problem: link

Solution: link

118: Power Walking

Problem: link

Tutorial: link

Solution: link

119: Hard Way

Problem: link

Tutorial: link

Solution: link

120: Madoka and Math Dad

Problem: link

Tutorial: link

Solution: link

121: Direction Change

Problem: link

Solution: link

122: Difference Operations

Problem: link

Solution: link

123: Ambitious Kid

Problem: link

Solution: link

124: MEX Repetition

Problem: link

Solution: link

125: Channel

Problem: link

Solution: link

126: Gift Carpet

Problem: link

Tutorial: link

Solution: link

127: Boxes Packing

Problem: link

Tutorial: link

Solution: link

128: The Modcrab

Problem: link

Tutorial: link

Solution: link

129: Hungry Student Problem

Problem: link

Tutorial: link

Solution: link

130: Find Extra One

Problem: link

Tutorial: link

Solution: link

sortings

131: Building an Aquarium

Problem: link

Solution: link

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157: Martian Dollar

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512: Geo Game

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513: Monocarp and the Set

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514: Salyg1n and Array (simple version)

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515: Replace With Product

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516: Guess Game

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517: The Great Equalizer

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518: String Mark

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519: Below the Diagonal

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520: The BOSS Can Count Pairs

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521: Red-Blue Operations (Easy Version)

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522: Collisions

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523: Longest Subsequence

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524: Intersection

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525: Equation

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526: Vlad and the Nice Paths (easy version)

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527: Another Wine Tasting Event

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528: Moving Dots

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530: Happy Sets

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531: Restoration of string

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532: More Wrong

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533: Bertown roads

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534: Rectangle Puzzle

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535: Area of Two Circles' Intersection

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536: Smart Boy

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537: Moon Craters

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538: What Has Dirichlet Got to Do with That?

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539: Ant on the Tree

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540: Thanos Nim

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541: Stripe 2

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542: Flag 2

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543: Start of the season

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544: The hat

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545: Lucky Sorting

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546: Work Group

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547: Graph Cost

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548: GCD Queries

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549: Carry Bit

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551: Wish I Knew How to Sort

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552: MEX vs MED

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553: Lights

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554: Swapping Characters

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555: Solitaire

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556: Safe

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557: Race

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558: Safe cracking

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559: Toys

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560: Multitest Generator

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561: Tree Master

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562: Serval and Shift-Shift-Shift

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563: Three Chairs

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564: Rectangle Shrinking

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565: Decomposition

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566: Kirill and Company

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567: Happy Life in University

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568: Yet Another Inversions Problem

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569: Light Bulbs (Hard Version)

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570: Sofia and Strings

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571: Infinite Card Game

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572: wxhtzdy ORO Tree

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573: Optimal Insertion

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574: Arithmetic Operations

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575: Exotic Queries

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576: Almost Difference

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577: Eyes Closed

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578: Ralph And His Tour in Binary Country

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579: Envy

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580: Trees and Segments

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581: Max to the Right of Min

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582: Competition

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583: Tenzing and Triangle

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584: Omsk Metro (hard version)

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585: MEX of LCM

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586: Fill the Matrix

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587: Lomsat gelral

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588: Ring Road 2

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589: Between

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590: Scheme

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591: Sum Over Zero

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592: Tokens on Graph

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593: Edge Reverse

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594: Small Permutation Problem (Easy Version)

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595: Compressed Tree

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596: Rubik's Cube Coloring (hard version)

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597: Another MEX Problem

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598: Imagination Castle

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599: Digital Wallet

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600: Non-Intersecting Subpermutations

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601: Balanced String

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602: Railguns

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603: Combinatorics Problem

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604: Petya and Coloring

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605: Nuclear Fusion

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606: Lesson Timetable

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607: Bath Queue

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608: Chain Chips

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609: Sequence

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610: Vlad and the Nice Paths (hard version)

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611: A Simple Task

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612: There Should Be a Lot of Maximums

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613: Minibuses on Venus (easy version)

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614: Sweets for Everyone!

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615: Bowls

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616: How Many Squares?

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617: Persistent Bookcase

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618: Symmetree

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619: Hospital Queue

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620: Monsters (hard version)

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621: Evaluate It and Back Again

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622: Hemose in ICPC ?

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623: Vika and Bonuses

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624: In Search of Truth (Easy Version)

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625: LuoTianyi and the Floating Islands (Hard Version)

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626: Strange town

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627: Magic Triples (Hard Version)

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628: Square Root of Permutation

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629: Berland Square

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630: Li Hua and Array

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631: Memory and Scores

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632: ZS and The Birthday Paradox

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633: City Union

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634: Explosions?

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635: Gaining Rating

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636: Node Pairs

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637: Valid Bitonic Permutations

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638: Game of the Year

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639: Partial Sorting

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640: Yet Another Array Counting Problem

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641: Salyg1n and Array (hard version)

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642: Playoff Fixing

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643: Sausage Maximization

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644: Cannon

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645: Hercule Poirot Problem

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646: Nearest vectors

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647: New Game with a Chess Piece

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648: Test

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649: Berland collider

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650: Deletion of Repeats

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651: Queue

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652: Vittorio Plays with LEGO Bricks

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653: Chemistry Lab

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654: Intersection and Union

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655: Vova Escapes the Matrix

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656: Removing Graph

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657: Balancing Weapons

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658: Survival of the Weakest (easy version)

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659: Routing

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660: Serval and Music Game

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661: Magician and Pigs (Easy Version)

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662: Rebrending

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663: Project Manager

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664: Josuke and Complete Graph

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665: Function Sum

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666: Hossam and a Letter

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667: Balance (Hard version)

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668: Interactive Game with Coloring

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669: Great Grids

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670: In Search of Truth (Hard Version)

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671: Aztec Catacombs

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672: Two Paths

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673: The Fox and the Complete Tree Traversal

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674: Square Table

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675: Multithreading

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676: Greedy Change

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677: Guess the String

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678: Inverse Transformation

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679: Xorcerer's Stones

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680: Cactus Wall

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681: Doremy's Experimental Tree

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682: Make It Connected

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683: The Beach

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684: Joking (Easy Version)

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685: Field Should Not Be Empty

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686: Babysitting

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687: Colorful Constructive

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688: Minimum Array

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689: Kuzya and Homework

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690: Frequency Queries

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691: Madoka and the Sixth-graders

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692: Willem, Chtholly and Seniorious

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693: Fast Travel Text Editor

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694: Rollbacks (Hard Version)

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695: Rollbacks (Easy Version)

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696: More Queries to Array...

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697: Editorial for Two

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698: Palindrome Partition

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699: Range Sorting (Hard Version)

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700: Walk the Runway

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701: LuoTianyi and XOR-Tree

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702: Frogs and mosquitoes

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703: Shooting Gallery

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704: Don't fear, DravDe is kind

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705: Construct Tree

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706: Small Permutation Problem (Hard Version)

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707: One-X

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708: Maximum And Queries (hard version)

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709: Fancy Arrays

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710: Vasya and Maximum Profit

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711: I Wanna be the Team Leader

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712: Axel and Marston in Bitland

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713: Korney Korneevich and XOR (hard version)

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714: Travel Plan

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715: Poachers

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716: Non-equal Neighbours

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717: Keen Tree Calculation

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718: Mighty Rock Tower

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719: Divide, XOR, and Conquer

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720: Strange Calculation and Cats

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721: Maximum Element

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722: Connecting Vertices

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723: Maximum Monogonosity

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724: Expected Destruction

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725: Alice and Recoloring 1

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726: Royal Questions

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727: Two Permutations (Easy Version)

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728: Tenzing and Tree

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730: Hyperregular Bracket Strings

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731: Population Size

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732: Red-Blue Operations (Hard Version)

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733: Trial for Chief

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734: King's Problem?

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735: Traveling in Berland

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736: Labeling the Tree with Distances

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737: Colored Subgraphs

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738: Spinach Pizza

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739: Library game

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740: Blocking Chips

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741: Velepin and Marketing

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742: XOR, Tree, and Queries

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743: Laboratory on Pluto

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744: Anya's Simultaneous Exhibition

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745: Construct Matrix

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746: Multiple Lamps

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747: Trees and XOR Queries Again

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748: Brukhovich and Exams

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749: Bored Bakry

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750: Ideal Farm

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751: Cutting Rectangle

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752: Array Equalizer

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753: Lihmuf Balling

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754: Ina of the Mountain

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755: The Boss's Identity

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756: Igor and Interesting Numbers

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757: Vika and Wiki

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758: Vika and Stone Skipping

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759: Min Cost Permutation (Easy Version)

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760: Boxes and Balls

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761: Lottery

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763: Typewriter

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765: Hypercatapult Commute

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772: Number Challenge

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773: Similar Polynomials

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774: Random Walk

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775: Emperor's Problem

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776: Number Table

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777: Interesting Sequence

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778: The Great Marathon

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779: Timber

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780: Testing

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781: Inverse Function

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783: Helper

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785: TV Game

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790: Tree

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791: Tetragon

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796: Festival Organization

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797: Graph Coloring (easy version)

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798: Graph Coloring (hard version)

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799: Majority

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801: Antifibonacci Cut

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802: Palindromic Problem

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804: Leha and security system

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805: BRT Contract

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806: Last Man Standing

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807: . . . Wait for it. . .

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808: XOR Partition

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809: Two Centroids

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810: LuoTianyi and the Function

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811: Cyclical Quest

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812: k-Maximum Subsequence Sum

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813: Tricky and Clever Password

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814: Li Hua and Path

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815: Points

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816: Holes

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817: M-tree

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818: Gasoline prices

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821: Weighed Tree Radius

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827: e-Government

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828: Miriany and Matchstick

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831: Sequence Transformation

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833: Forward, march!

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834: Minibuses on Venus (hard version)

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837: Another n-dimensional chocolate bar

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838: Halve or Subtract

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839: Serval and Brain Power

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841: Edge Queries

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843: Alice and Recoloring 2

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844: Most Different Tree

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845: Flower-like Pseudotree

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847: GCD Master (hard version)

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848: GCD Master (easy version)

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849: Teamwork

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850: Wonderful Jump

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851: Matrix Rank (Hard Version)

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852: Matrix Rank (Easy Version)

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853: Split

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854: Hard Design

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855: Ksusha and Square

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856: Lazy Numbers

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857: Swaps

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858: Nephren Runs a Cinema

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859: Lust

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860: Evaluate RBS

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861: Mod Mod Mod

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862: PermuTree (hard version)

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863: Game Bundles

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864: Triangle Platinum?

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865: Tree Weights

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866: Min Cost Permutation (Hard Version)

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867: Swimmers in the Pool

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868: Tenzing and Random Operations

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869: Tenzing and Random Real Numbers

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870: Doctor's Brown Hypothesis

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871: Pumping Lemma

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872: Landscaping

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873: Symmetric Projections

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874: Monocarp and a Strategic Game

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875: Stuck Conveyor

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876: Spider

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877: Fading into Fog

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878: Graph Game

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879: Toy Machine

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880: Petya and Rectangle

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881: Edge coloring of bipartite graph

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882: Cut Length

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883: Parquet Re-laying

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884: DravDe saves the world

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885: Palisection

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886: Communication Towers

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887: Approximate Diameter

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888: Strange Triples

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889: Removal Sequences

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890: Hero to Zero

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891: Infinite Chess

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892: The Game of the Century

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893: MCF

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894: Doremy's Perfect DS Class (Medium Version)

Problem: link

Solution: link

895: Doremy's Perfect DS Class (Easy Version)

Problem: link

Solution: link

896: Decent Division

Problem: link

Solution: link

898: Location

Problem: link

Solution: link

899: Kazaee

Problem: link

Solution: link

trees

900: Michael and Hotel

Problem: link

Solution: link

901: Baldman and the military

Problem: link

Tutorial: link

Solution: link

902: Bracket Insertion

Problem: link

Tutorial: link

Solution: link

903: Distance to the Path

Problem: link

Tutorial: link

Solution: link

black

constructive algorithms

904: Minimum Segments

Problem: link

Tutorial: link

Solution: link

905: Good Colorings

Problem: link

Tutorial: link

Solution: link

906: Two Permutations (Hard Version)

Problem: link

Solution: link

907: Magic Square

Problem: link

Solution: link

908: Bus Routes

Problem: link

Solution: link

909: OH NO1 (-2-3-4)

Problem: link

Solution: link

910: Maximum Permutation

Problem: link

Solution: link

911: Koxia, Mahiru and Winter Festival

Problem: link

Solution: link

912: Diverse Coloring

Problem: link

Tutorial: link

Solution: link

913: Unequal Adjacent Elements

Problem: link

Solution: link

914: Dangerous Laser Power

Problem: link

Tutorial: link

Solution: link

data structures

915: Parallel Swaps Sort

Problem: link

Solution: link

916: Diamond Theft

Problem: link

Tutorial: link

Solution: link

917: Clubstep

Problem: link

Solution: link

918: Standard Graph Problem

Problem: link

Tutorial: link

Solution: link

919: MEXanization

Problem: link

Tutorial: link

Solution: link

920: Freak Joker Process

Problem: link

Solution: link

921: Welcome home, Chtholly

Problem: link

Solution: link

922: Rivalries

Problem: link

Tutorial: link

Solution: link

923: LuoTianyi and Cartridge

Problem: link

Solution: link

924: A task for substrings

Problem: link

Solution: link

925: Iron Man

Problem: link

Tutorial: link

Solution: link

926: Flow Control

Problem: link

Solution: link

927: Treasure Hunt

Problem: link

Solution: link

928: Codeforces Scoreboard

Problem: link

Solution: link

929: Two Subtrees

Problem: link

Tutorial: link

Solution: link

930: Doremy's Paint 2

Problem: link

Solution: link

931: MEX Tree Manipulation

Problem: link

Tutorial: link

Solution: link

dp

932: Optimizations From Chelsu

Problem: link

Tutorial: link

Solution: link

933: Ball-Stackable

Problem: link

Solution: link

934: Jellyfish and Inscryption

Problem: link

Solution: link

935: Min-Sum-Max

Problem: link

Solution: link

936: Redundant Routes

Problem: link

Solution: link

937: Goldberg Machine 3

Problem: link

Solution: link

938: Asterism Stream

Problem: link

Solution: link

939: Sloth

Problem: link

Solution: link

940: Panda Meetups

Problem: link

Tutorial: link

Solution: link

941: Old Mobile

Problem: link

Solution: link

942: Zombies

Problem: link

Tutorial: link

Solution: link

943: Deja Vu

Problem: link

Tutorial: link

Solution: link

944: Willy-nilly, Crack, Into Release!

Problem: link

Solution: link

945: Bosco and Particle

Problem: link

Solution: link

946: The Maximum Prefix

Problem: link

Solution: link

947: Code Lock

Problem: link

Solution: link

948: Tree Cutting

Problem: link

Solution: link

949: Crossing the Railways

Problem: link

Solution: link

950: Segment Covering

Problem: link

Solution: link

951: Infinite Game

Problem: link

Tutorial: link

Solution: link

greedy

952: Half-sum

Problem: link

Solution: link

953: N Machines

Problem: link

Tutorial: link

Solution: link

math

954: Fugitive Frenzy

Problem: link

Tutorial: link

Solution: link

955: Parallel Universes (Hard)

Problem: link

Solution: link

956: Survival of the Weakest (hard version)

Problem: link

Tutorial: link

Solution: link

957: Window Signals (easy version)

Problem: link

Tutorial: link

Solution: link

958: Koxia and Sequence

Problem: link

Solution: link

959: Koxia and Bracket

Problem: link

Solution: link

960: Anti-median (Hard Version)

Problem: link

Solution: link

961: Anti-median (Easy Version)

Problem: link

Solution: link

misc

962: Indefinite Clownfish

Problem: link

Solution: link

963: Minimums or Medians

Problem: link

Tutorial: link

Solution: link

964: Olympic Team Building

Problem: link

Solution: link

965: Doremy's Perfect DS Class (Hard Version)

Problem: link

Solution: link

966: Minecraft Series

Problem: link

Tutorial: link

Solution: link

967: Joking (Hard Version)

Problem: link

Solution: link

trees

968: Roads in E City

Problem: link

Solution: link

Appendix: some common template code

MInt and MLong

The M in the struct name is for "modular arithmetic". Whenever a problem asks for "submit answer modulo 1e9+7 or 998244353" , this is the data structure to use. This is the most popular template by far, appearing in ~10% of problems.

```
struct MInt {
    int x;
    constexpr MInt() : x{} {}
    constexpr MInt(i64 x) : x{norm(x % P)} {}

    constexpr int norm(int x) const {
        if (x < 0) {
            x += P;
        }
        if (x >= P) {
            x -= P;
        }
        return x;
    }
    constexpr int val() const {
        return x;
    }
    explicit constexpr operator int() const {
        return x;
    }
    constexpr MInt operator-() const {
        MInt res;
        res.x = norm(P - x);
        return res;
    }
    constexpr MInt inv() const {
        assert(x != 0);
        return power(*this, P - 2);
    }
```

```
30      constexpr MInt &operator*=(MInt rhs) {
31          x = 1LL * x * rhs.x % P;
32          return *this;
33      }
34      constexpr MInt &operator+=(MInt rhs) {
35          x = norm(x + rhs.x);
36          return *this;
37      }
38      constexpr MInt &operator-=(MInt rhs) {
39          x = norm(x - rhs.x);
40          return *this;
41      }
42      constexpr MInt &operator/=(MInt rhs) {
43          return *this *= rhs.inv();
44      }
45      friend constexpr MInt operator*(MInt lhs, MInt rhs) {
46          MInt res = lhs;
47          res *= rhs;
48          return res;
49      }
50      friend constexpr MInt operator+(MInt lhs, MInt rhs) {
51          MInt res = lhs;
52          res += rhs;
53          return res;
54      }
55      friend constexpr MInt operator-(MInt lhs, MInt rhs) {
56          MInt res = lhs;
57          res -= rhs;
58          return res;
59      }
60      friend constexpr MInt operator/(MInt lhs, MInt rhs) {
61          MInt res = lhs;
62          res /= rhs;
63          return res;
64      }
65      friend constexpr istream &operator>>(istream &is, MInt &a) {
66          i64 v;
67          is >> v;
68          a = MInt(v);
69          return is;
70      }
71      friend constexpr ostream &operator<<(ostream &os, const MInt &a) {
72          return os << a.val();
73      }
74      friend constexpr bool operator==(MInt lhs, MInt rhs) {
75          return lhs.val() == rhs.val();
76      }
77      friend constexpr bool operator!=(MInt lhs, MInt rhs) {
78          return lhs.val() != rhs.val();
79      }
80  }
81
82  struct MLong {
83      i64 x;
84      constexpr MLong() : x{} {}
85      constexpr MLong(i64 x) : x{norm(x % getMod())} {}
86
87      static i64 Mod;
88      constexpr static i64 getMod() {
89          if (P > 0) {
90              return P;
91          } else {
92              return Mod;
93          }
```

```
94      }
95      constexpr static void setMod(i64 Mod_) {
96          Mod = Mod_;
97      }
98      constexpr i64 norm(i64 x) const {
99          if (x < 0) {
100             x += getMod();
101         }
102         if (x >= getMod()) {
103             x -= getMod();
104         }
105         return x;
106     }
107     constexpr i64 val() const {
108         return x;
109     }
110     explicit constexpr operator i64() const {
111         return x;
112     }
113     constexpr MLong operator-() const {
114         MLong res;
115         res.x = norm(getMod() - x);
116         return res;
117     }
118     constexpr MLong inv() const {
119         assert(x != 0);
120         return power(*this, getMod() - 2);
121     }
122     constexpr MLong &operator*=(MLong rhs) & {
123         x = mul(x, rhs.x, getMod());
124         return *this;
125     }
126     constexpr MLong &operator+=(MLong rhs) & {
127         x = norm(x + rhs.x);
128         return *this;
129     }
130     constexpr MLong &operator-=(MLong rhs) & {
131         x = norm(x - rhs.x);
132         return *this;
133     }
134     constexpr MLong &operator/=(MLong rhs) & {
135         return *this *= rhs.inv();
136     }
137     friend constexpr MLong operator*(MLong lhs, MLong rhs) {
138         MLong res = lhs;
139         res *= rhs;
140         return res;
141     }
142     friend constexpr MLong operator+(MLong lhs, MLong rhs) {
143         MLong res = lhs;
144         res += rhs;
145         return res;
146     }
147     friend constexpr MLong operator-(MLong lhs, MLong rhs) {
148         MLong res = lhs;
149         res -= rhs;
150         return res;
151     }
152     friend constexpr MLong operator/(MLong lhs, MLong rhs) {
153         MLong res = lhs;
154         res /= rhs;
155         return res;
156     }
157     friend constexpr istream &operator>>(istream &is, MLong &a) {
```

```
158         i64 v;
159         is >> v;
160         a = MLong(v);
161         return is;
162     }
163     friend constexpr ostream &operator<<(ostream &os, const MLong &a) {
164         return os << a.val();
165     }
166     friend constexpr bool operator==(MLong lhs, MLong rhs) {
167         return lhs.val() == rhs.val();
168     }
169     friend constexpr bool operator!=(MLong lhs, MLong rhs) {
170         return lhs.val() != rhs.val();
171     }
172 }
```

Comb

For combinatorial math like binomials.

```
1  struct Comb {
2      int n;
3      vector<Z> _fac;
4      vector<Z> _invfac;
5      vector<Z> _inv;
6
7      Comb() : n{0}, _fac{1}, _invfac{1}, _inv{0} {}
8      Comb(int n) : Comb() {
9          init(n);
10     }
11
12     void init(int m) {
13         if (m <= n) return;
14         _fac.resize(m + 1);
15         _invfac.resize(m + 1);
16         _inv.resize(m + 1);
17
18         for (int i = n + 1; i <= m; i++) {
19             _fac[i] = _fac[i - 1] * i;
20         }
21         _invfac[m] = _fac[m].inv();
22         for (int i = m; i > n; i--) {
23             _invfac[i - 1] = _invfac[i] * i;
24             _inv[i] = _invfac[i] * _fac[i - 1];
25         }
26         n = m;
27     }
28
29     Z fac(int m) {
30         if (m > n) init(2 * m);
31         return _fac[m];
32     }
33     Z invfac(int m) {
34         if (m > n) init(2 * m);
35         return _invfac[m];
36     }
37     Z inv(int m) {
38         if (m > n) init(2 * m);
39         return _inv[m];
40     }
41     Z binom(int n, int m) {
```

```
        if (n < m || m < 0) return 0;
        return fac(n) * invfac(m) * invfac(n - m);
    }
}
```

DSU

The short implementation of DSU, also known as “union-find”.

```
struct DSU {
    vector<int> f, siz;

    DSU() {}
    DSU(int n) {
        init(n);
    }

    void init(int n) {
        f.resize(n);
        iota(f.begin(), f.end(), 0);
        siz.assign(n, 1);
    }

    int find(int x) {
        while (x != f[x]) {
            x = f[x] = f[f[x]];
        }
        return x;
    }

    bool same(int x, int y) {
        return find(x) == find(y);
    }

    bool merge(int x, int y) {
        x = find(x);
        y = find(y);
        if (x == y) {
            return false;
        }
        siz[x] += siz[y];
        f[y] = x;
        return true;
    }

    int size(int x) {
        return siz[find(x)];
    }
}
```

Fenwick

Fenwick is also known as a Binary Index Tree.

```
struct Fenwick {
    int n;
    vector<T> a;
```

```

    Fenwick(int n = 0) {
        init(n);
    }

    void init(int n) {
        this->n = n;
        a.assign(n, T());
    }

    void add(int x, T v) {
        for (int i = x + 1; i <= n; i += i & -i) {
            a[i - 1] += v;
        }
    }

    T sum(int x) {
        auto ans = T();
        for (int i = x; i > 0; i -= i & -i) {
            ans += a[i - 1];
        }
        return ans;
    }

    T rangeSum(int l, int r) {
        return sum(r) - sum(l);
    }

    int kth(T k) {
        int x = 0;
        for (int i = 1 << __lg(n); i; i /= 2) {
            if (x + i <= n && k >= a[x + i - 1]) {
                x += i;
                k -= a[x - 1];
            }
        }
        return x;
    }
}
```

HLD

HLD

```
struct HLD {
    int n;
    vector<int> siz, top, dep, parent, in, out, seq;
    vector<vector<int>> adj;
    int cur;

    HLD() {}
    HLD(int n) {
        init(n);
    }
    void init(int n) {
        this->n = n;
        siz.resize(n);
        top.resize(n);
        dep.resize(n);
        parent.resize(n);
        in.resize(n);
```

```
        out.resize(n);
        seq.resize(n);
        cur = 0;
        adj.assign(n, {});
    }
    void addEdge(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }
    void work(int root = 0) {
        top[root] = root;
        dep[root] = 0;
        parent[root] = -1;
        dfs1(root);
        dfs2(root);
    }
    void dfs1(int u) {
        if (parent[u] != -1) {
            adj[u].erase(find(adj[u].begin(), adj[u].end(), parent[u]));
        }

        siz[u] = 1;
        for (auto &v : adj[u]) {
            parent[v] = u;
            dep[v] = dep[u] + 1;
            dfs1(v);
            siz[u] += siz[v];
            if (siz[v] > siz[adj[u][0]]) {
                swap(v, adj[u][0]);
            }
        }
    }
    void dfs2(int u) {
        in[u] = cur++;
        seq[in[u]] = u;
        for (auto v : adj[u]) {
            top[v] = v == adj[u][0] ? top[u] : v;
            dfs2(v);
        }
        out[u] = cur;
    }
    int lca(int u, int v) {
        while (top[u] != top[v]) {
            if (dep[top[u]] > dep[top[v]]) {
                u = parent[top[u]];
            } else {
                v = parent[top[v]];
            }
        }
        return dep[u] < dep[v] ? u : v;
    }

    int dist(int u, int v) {
        return dep[u] + dep[v] - 2 * dep[lca(u, v)];
    }

    int jump(int u, int k) {
        if (dep[u] < k) {
            return -1;
        }

        int d = dep[u] - k;

        while (dep[top[u]] > d) {
```

```
            u = parent[top[u]];
        }

        return seq[in[u] - dep[u] + d];
    }

    bool isAncester(int u, int v) {
        return in[u] <= in[v] && in[v] < out[u];
    }

    int rootedParent(int u, int v) {
        swap(u, v);
        if (u == v) {
            return u;
        }
        if (!isAncester(u, v)) {
            return parent[u];
        }
        auto it = upper_bound(adj[u].begin(), adj[u].end(), v, [&](int x, int y) {
            return in[x] < in[y];
        }) - 1;
        return *it;
    }

    int rootedSize(int u, int v) {
        if (u == v) {
            return n;
        }
        if (!isAncester(v, u)) {
            return siz[v];
        }
        return n - siz[rootedParent(u, v)];
    }

    int rootedLca(int a, int b, int c) {
        return lca(a, b) ^ lca(b, c) ^ lca(c, a);
    }
}
```

LazySegmentTree

Lazy Segment Tree

```
struct LazySegmentTree {
    int n;
    vector<Info> info;
    vector<Tag> tag;
    LazySegmentTree() : n(0) {}
    LazySegmentTree(int n_, Info v_ = Info()) {
        init(n_, v_);
    }
    template<class T>
    LazySegmentTree(vector<T> init_) {
        init(init_);
    }
    void init(int n_, Info v_ = Info()) {
        init(vector(n_, v_));
    }
    template<class T>
    void init(vector<T> init_) {
        n = init_.size();
```

```
        info.assign(4 << __lg(n), Info());
        tag.assign(4 << __lg(n), Tag());
        function<void(int, int, int)> build = [&](int p, int l, int r) {
            if (r - l == 1) {
                info[p] = init_[l];
                return;
            }
            int m = (l + r) / 2;
            build(2 * p, l, m);
            build(2 * p + 1, m, r);
            pull(p);
        };
        build(1, 0, n);
    }
    void pull(int p) {
        info[p] = info[2 * p] + info[2 * p + 1];
    }
    void apply(int p, const Tag &v) {
        info[p].apply(v);
        tag[p].apply(v);
    }
    void push(int p) {
        apply(2 * p, tag[p]);
        apply(2 * p + 1, tag[p]);
        tag[p] = Tag();
    }
    void modify(int p, int l, int r, int x, const Info &v) {
        if (r - l == 1) {
            info[p] = v;
            return;
        }
        int m = (l + r) / 2;
        push(p);
        if (x < m) {
            modify(2 * p, l, m, x, v);
        } else {
            modify(2 * p + 1, m, r, x, v);
        }
        pull(p);
    }
    void modify(int p, const Info &v) {
        modify(1, 0, n, p, v);
    }
    Info rangeQuery(int p, int l, int r, int x, int y) {
        if (l >= y || r <= x) {
            return Info();
        }
        if (l >= x && r <= y) {
            return info[p];
        }
        int m = (l + r) / 2;
        push(p);
        return rangeQuery(2 * p, l, m, x, y) + rangeQuery(2 * p + 1, m, r, x, y);
    }
    Info rangeQuery(int l, int r) {
        return rangeQuery(1, 0, n, l, r);
    }
    void rangeApply(int p, int l, int r, int x, int y, const Tag &v) {
        if (l >= y || r <= x) {
            return;
        }
        if (l >= x && r <= y) {
            apply(p, v);
            return;
```

```
83          }
84          int m = (l + r) / 2;
85          push(p);
86          rangeApply(2 * p, l, m, x, y, v);
87          rangeApply(2 * p + 1, m, r, x, y, v);
88          pull(p);
89      }
90      void rangeApply(int l, int r, const Tag &v) {
91          return rangeApply(1, 0, n, l, r, v);
92      }
93      template<class F>
94      int findFirst(int p, int l, int r, int x, int y, F pred) {
95          if (l >= y || r <= x || !pred(info[p])) {
96              return -1;
97          }
98          if (r - l == 1) {
99              return l;
100         }
101         int m = (l + r) / 2;
102         push(p);
103         int res = findFirst(2 * p, l, m, x, y, pred);
104         if (res == -1) {
105             res = findFirst(2 * p + 1, m, r, x, y, pred);
106         }
107         return res;
108     }
109     template<class F>
110     int findFirst(int l, int r, F pred) {
111         return findFirst(1, 0, n, l, r, pred);
112     }
113     template<class F>
114     int findLast(int p, int l, int r, int x, int y, F pred) {
115         if (l >= y || r <= x || !pred(info[p])) {
116             return -1;
117         }
118         if (r - l == 1) {
119             return l;
120         }
121         int m = (l + r) / 2;
122         push(p);
123         int res = findLast(2 * p + 1, m, r, x, y, pred);
124         if (res == -1) {
125             res = findLast(2 * p, l, m, x, y, pred);
126         }
127         return res;
128     }
129     template<class F>
130     int findLast(int l, int r, F pred) {
131         return findLast(1, 0, n, l, r, pred);
132     }
133 }
```